

KUNAL VAGHANI

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AI/ML & Data Science Student | Machine Learning, LLM Systems, Information Retrieval, Python & C++ Engineering

SUMMARY

AI & Data Science student and software engineer building inspectable, hardware-aware **AI systems**, including local **LLM** inference pipelines, **agent runtimes**, and **information retrieval** systems. Strong foundation in **Python** and **C++** performance engineering, with hands-on experience in **machine learning**, evaluation, and reproducible benchmarking on resource-constrained hardware. Seeking an **AI/ML** or **Data Science** internship to apply systems-level engineering skills to real-world machine learning problems.

EDUCATION

Parul University	2026 - 2030
B.Tech, Artificial Intelligence and Data Science	
Humber College	Jan 2022 - Aug 2024
Advanced Diploma, Game Programming — Dean's Honor Roll, Faculty of Business (Winter 2023)	

TECHNICAL SKILLS

- Languages:** Python, C++, SQL, TypeScript
- Core CS:** Data Structures, Algorithms, Software Design, System Design, Object-Oriented Design
- AI/ML & Data Science:** Machine Learning, Natural Language Processing, Local LLM Inference (llama.cpp), Agent Runtimes, Model Routing, Evaluation, Embeddings
- Information Retrieval:** BM25, Inverted Indexes, Vector Search, Hybrid Retrieval, Reranking, Multilingual Retrieval
- Systems & Performance:** Concurrency, Memory Management, Profiling, Latency/Throughput Optimization, Linux, Access Control & Security
- Web & Data Tooling:** React, SQLite, Git/GitHub, Software Testing, Reproducibility, Observability
- Certifications:** Intro to Machine Learning (Kaggle, 2026); Develop a Generative AI Chat App Using the Microsoft Foundry SDK (Microsoft Applied Skills, 2026); Enhance Agents with Autonomous Capabilities (Microsoft Applied Skills, 2026)

PROJECTS

- Local AI Systems Lab — Local LLM Runtime and Agent Infrastructure** GitHub
- Designed the **system architecture** for an inspectable local AI runtime in **Python** and **C++**, applying **software design** and **data structures** (state machines, priority queues, hash-chained logs) to build an explicit agent state machine, bounded priority scheduler, and model registry with explained routing logic.
 - Implemented deterministic replay, hash-chained execution traces, and an **access-control** layer with guarded, exact-grant tool permissions, alongside **SQLite**-based checkpoint/recovery for reliability under fault injection.
 - Integrated real local **LLM inference** using **llama.cpp** on constrained hardware (RTX 3050, 4GB VRAM), avoiding paid inference APIs.
 - Built a **React/TypeScript** frontend and validated the full system end-to-end with 154 backend **tests** and 39 frontend tests passing, documenting acceptance evidence for each release stage.
 - Tech: Python, TypeScript, React, SQLite, llama.cpp, SSE, C++ runtime
- Local LLM Performance Lab — Hardware-Aware Inference Benchmarking** GitHub
- Designed a **benchmarking** track to measure **latency**, memory, and failure behavior of quantized **LLM inference** under a strict 4GB VRAM budget.
 - Built live hardware **profiling**, an admission policy, a model/profile registry, and GPU/CPU offload controls with automated regression gates.
 - Measured and documented real performance data on a Qwen2.5 GGUF model: 1,801 ms time-to-first-token, 103.5 tokens/sec generation, and 1,189 MiB peak VRAM delta.
 - Tech: llama.cpp, Qwen2.5, GGUF, CUDA, Python, Hardware Telemetry
- Multilingual Retrieval Engine — NLP & Information Retrieval System (English, Hindi, Gujarati)**
- Architected a local, multilingual **natural language processing** and **information retrieval** pipeline combining lexical (**BM25**), dense vector, and hybrid score-fusion retrieval, using **algorithmic** tokenization and inverted-index **data structures**.
 - Designed **evaluation** methodology using Recall, MRR, and NDCG to independently validate retrieval and reranking quality before generation.
 - Tech: Python, C++, NLP, BM25, Vector Search, Hybrid Retrieval, Evaluation Metrics

SYSTEMS FOUNDATION

Prior **C++** engineering background in real-time systems and graphics (custom OpenGL/Vulkan game engines, GPU resource management) developed the low-level **performance** discipline now applied to AI systems work. Full details: github.com/kunalvaghani/Kunals_Repo

ADDITIONAL EXPERIENCE

- Team Lead — Tim Hortons** Mar 2022 - Sep 2023
- Trained team members and coordinated back-office reporting and reconciliation, building organizational and communication skills applicable to collaborative engineering teams.
- Sales Associate — Eataly Sherway** Oct 2023 - 2024
- Managed inventory and customer service in a fast-paced retail environment, strengthening communication and attention to detail.